# HongFei Huang

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## Skills

**Languages:** C++, C#, C, Python
**XR & Graphics:** Unity, OpenXR, Meta Quest, OpenGL, OpenUSD
**Tools & Areas:** Linux, Git, Machine Learning, PyTorch, AI Coding Harness, Game Engines, XR Input/Output, Rendering Pipelines

## Experience

**SideFX** **Sep 2025 – Apr 2026**
*3D Software Developer, Internship* (C++ | Python | OpenUSD | Linux) *Toronto, ON*

- Shipped a USD prim scanner that auto-patches missing material bindings at zero runtime overhead, raising material consistency across shots by **30%**.
- Built primitive coordinate conversion for 4×4 matrices, preserving local transforms via Xform Common API decomposition (pivot, rotation, shear); raised successful Maya–Houdini transform round-trips by **60%**.
- Reduced "Clean All Transforms" lookup complexity from $O(N^2)$ to $O(N)$ in the transform panel.

**Waterloo Reality Labs** **Sep 2024 – Aug 2025**
*VR Software Engineer & Co-Lead, Part-time* (Unity | C# | OpenXR | PyTorch) *Waterloo, ON*

- Developed an **OpenXR-based Unity hand-tracking pipeline** for collecting temporal gesture sequences and training dynamic gesture-recognition models.
- Implemented a **Conv2D model** and adaptive preprocessing pipeline for Unity Barracuda, including automated CLI tests and ONNX export.

**Huawei** **Sep 2023 – Dec 2023**
*Software Engineer, Internship* (RTOS | C | Python | CI/CD) *Kanata, ON*

- Built a user-space validation tool for RTOS priority boosting, enabling CI to catch priority-inversion regressions.
- Automated OS metric analysis in Python (NumPy, Matplotlib), improving performance/power profiling by **60%**.

**OpenText** **Jan 2023 – Apr 2023**
*Software Developer, Internship* (OScript | Profiling | Linux) *Waterloo, ON*

- Profiled OScript data-processing workloads on Red Hat Linux to identify performance bottlenecks.

**OpenText** **Jan 2022 – Apr 2022**
*Software Developer, Internship* (C++ | Language Server | Linux) *Waterloo, ON*

- Prototyped a C++ language server on Red Hat Linux for compiler-aware editor tooling.

## Projects

**VR Geometric Algebra Viewer** (C# | Unity | Meta Quest) **Sep 2024 – Jul 2026**

- Built a mixed-reality geometric algebra tool on Meta Quest 3 with **hand-tracked geometry construction** and previews of rotations, translations, and screw motions.
- Implemented an **ANTLR expression interpreter** with snapshot and live assignments, sharing object creation between console input and hand-driven construction.
- Reduced average frame time by **60%** in static scenes with **20 live bindings** on Meta Quest 3 by skipping geometry recomputation when inputs remained unchanged.

**Real-Time Rigid Body Fracture Simulation** (C++ | OpenGL | Computer Graphics) **Mar 2023 – Apr 2023**

- Engineered an interactive **OpenGL** fracture simulation using pre-fractured 3D models.
- Implemented a Surface Area Heuristic Bounding Volume Hierarchy **(SAH-BVH)** for Ray-AABB and AABB-AABB collision detection, sustaining 170+ FPS with complex inputs.

## Education

**University of Waterloo**, Master of Mathematics in Computer Science, Co-op **Sep 2024 – Aug 2026**
**University of Waterloo**, Bachelor of Applied Science in Computer Engineering, Co-op **Sep 2019 – Apr 2024**